



Certified Penetration Testing Professional

Nmap Cheat Sheet

Nmap

Source: <https://nmap.org>

Nmap is a security scanner for network exploration and hacking. It allows you to discover hosts and services on a computer network, thus creating a "map" of the network. It sends specially crafted packets to the target host and then analyzes the responses to accomplish its goal. Either a network administrator or an attacker can use this tool for their specific needs.

Syntax

`nmap [Scan Type...] [Options] {Target specification}`

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1. Nmap Options

Option (Switch/ Syntax)	Description
Target Specification	
<code>-iL <inputfilename></code>	Input from list of hosts/networks
<code>-iR <num hosts></code>	Choose random targets/ Scan random hosts <code>nmap -iR [number]</code>
<code>--exclude <host1[,host2][,host3],...></code>	Exclude single or multiple hosts/networks

<code>--excludefile <exclude_file></code>	Exclude list from file
Host Discovery	
<code>-sL</code>	List Scan - simply lists targets <code>nmap <Target IP>-3 -sL</code>
<code>-sn</code>	Ping Scan - disable port scan for discovering host <code>nmap <Target IP>/24 -sn</code>
<code>-Pn</code>	Treat all hosts as online -- skip host discovery <code>nmap <Target IP>-5 -Pn</code>
<code>-PS/PA/PU/PY[portlist]</code>	TCP SYN/ACK, UDP or SCTP INIT discovery to given ports
<code>-PE/PP/PM</code>	ICMP echo, timestamp, and netmask request discovery probes
<code>-PP</code>	Use ICMP timestamp request
<code>-PO[protocol list]</code>	IP Protocol Ping
<code>-n/-R</code>	Never do DNS resolution/Always resolve [default: sometimes] <code>nmap -n <Target IP></code> <code>nmap -R <Target IP></code>
<code>--dns-servers <serv1[,serv2],...></code>	Specify custom DNS servers <code>nmap --dns-servers server1 server2 <Target IP></code>
<code>--system-dns</code>	Use OS's DNS resolver <code>nmap --system-dns <Target IP></code>
<code>--traceroute</code>	Trace hop path to each host <code>nmap --traceroute <Target IP></code>
<code>-PR</code>	ARP discovery on local network <code>nmap <Target IP>-1/24 -PR</code>

Scan Techniques	
<code>-sS/sT/sA/sW/sM</code>	TCP SYN/Connect()/ACK/Window/Maimon scans
<code>-sU</code>	UDP Scan <code>nmap -sU -v <Target IP></code> UDP port scan <code>nmap <Target IP> -sU</code>
<code>-sN/sF/sX</code>	TCP Null, FIN, and Xmas scans
<code>-scanflags=value -sA</code>	TCP ACK scan <code>nmap -scanflags=value -sA <Target IP></code>
<code>-scanflags</code>	TCP scan flags <code>nmap --scanflags <Target IP></code>
<code>-Sp</code>	Ping scan <code>nmap -Sp <Target IP></code>
<code>--scanflags <flags></code>	Customize TCP scan flags
<code>-sI <zombie host[:probeport]></code>	Idle zombie scan <code>nmap -sI zombie <Target IP></code>
<code>-sY/sZ</code>	SCTP INIT scan <code>nmap -sY -v <Target IP></code> SCTP COOKIE-ECHO scan <code>nmap -sZ -v <Target IP></code>
<code>-sO</code>	IP protocol scan <code>nmap -sO <Target IP></code>
<code>-b <FTP relay host></code>	FTP bounce scan
<code>-send-eth</code>	Send raw ethernet packets <code>nmap -send-eth <Target IP></code>
<code>-send-ip</code>	Send IP packets <code>nmap -send-ip <Target IP></code>
Port Specification and Scan Order	
<code>-p <port ranges></code>	Only scan specified range ports <code>nmap -p 1-100 <Target IP></code> e.g. <code>-p80,443</code> or <code>-p1-65535</code>

<code>-p-</code>	Port scans all 1-65535 ports <code>nmap <Target IP> -p-</code>
<code>-p <protocol></code>	Port scan from specified protocols <code>nmap -smtp,https <Target IP></code>
<code>-F</code>	Fast mode - Scan less ports than the default scan (scan 100 most common ports) <code>nmap <Target IP> -F</code>
<code>-r</code>	Scan ports consecutively – do not randomize
<code>-randomize-hosts</code>	Randomize target host order <code>nmap -randomize-hosts <Target IP></code>
<code>-p<port1>,<port2>,...</code>	Port list
<code>-p<port1>-<port2></code>	Port range
<code>-p "*"</code>	Scan port using name <code>nmap -p "*" ftp <Target IP></code>
<code>-pU:53,U:110,T20-445</code>	Mix TCP and UDP
<code>--top-ports <number></code>	Scan <number> most common ports
<code>--port-ratio <ratio></code>	Scan ports more common than <ratio>
<code>-p-65535</code>	Leaving off initial port in range makes Nmap scan start at port 1 <code>nmap <Target IP> -p-65535</code> Leaving off initial port in range makes the scan start at port 1 <code>nmap -p-65535 <Target IP></code>
<code>-p0-</code>	Leaving off end port in range makes Nmap scan through port 65535 <code>nmap <Target IP> -p0-</code> <code>nmap -p0- <Target IP></code>
Service/Version Detection	
<code>-sV</code>	Probe open ports to determine service/version info <code>nmap <Target IP> -sV</code>
<code>--version-intensity <level></code>	Set from 0 (light) to 9 (try all probes)

<code>--version-light</code>	Limit to most likely probes (intensity 2)
<code>--version-all</code>	Try every single probe (intensity 9)
<code>--version-trace</code>	Show detailed version scan activity (for debugging)
Script Scan	
<code>--script=<ScriptName> <ScriptCategory> <ScriptDir>...</code>	Run individual or group of scripts
<code>--script=<Lua scripts></code>	<Lua scripts> is a comma separated list of directories, script-files or script-categories
<code>--script-args=<n1=v1,[n2=v2,...]></code>	Provide arguments to scripts
<code>--script-args-file=filename</code>	provide NSE script args in a file
<code>--script-trace</code>	Show all data sent and received
<code>--script-updatedb</code>	Update the script database. <code>nmap --script-updatedb</code>
<code>--script-help</code>	"Lua scripts" = Show help about scripts
OS Detection	
<code>-O</code>	Enable OS detection/ OS Discovery using Nmap and Unicornscan/ Remote OS Detection using TCP/IP stack fingerprinting <code>nmap -O <Target IP></code>
<code>--osscan-limit</code>	Limit OS detection to promising targets
<code>--osscan-guess</code>	Guess OS more aggressively
<code>--max-os-tries</code>	Set the maximum number x of OS detection tries against a target
Timing and Performance	
<code>-T<0-5></code>	Set timing template (higher is faster)
<code>-ttl [time]</code>	Set the packet TTL <code>nmap -ttl [time] <Target IP></code>
<code>--min-hostgroup/max-hostgroup <size></code>	Parallel host scan group sizes

<code>--min-parallelism/max-parallelism <numprobes></code>	Probe parallelization
<code>--min-rtt-timeout/max-rtt-timeout/initial-rtt-timeout <time></code>	Specifies probe round trip time
<code>--max-retries <tries></code>	Caps number of port scan probe retransmissions
<code>--host-timeout <time></code>	Give up on target after this long
<code>--scan-delay/--max-scan-delay <time></code>	Adjust delay between probes
<code>--min-rate <number></code>	Send packets no slower than <number> per second
<code>--max-rate <number></code>	Send packets no faster than <number> per second
<code>-defeat-rst-ratelimit</code>	Defeat reset rate limits <code>nmap -defeat-rst-ratelimit <Target IP></code>
Firewall/IDS Evasion and Spoofing	
<code>-f; --mtu <val></code>	Fragment packets (optionally w/given MTU)
<code>-D <decoy1,decoy2[,ME],...></code>	Cloak a scan with decoys
<code>-S <IP_Address></code>	Spoof source address
<code>-e <iface></code>	Use specified interface <code>nmap -e [interface] <Target IP></code>
<code>-g/--source-port <portnum></code>	Use given port number
<code>--data-length <num></code>	Append random data to send packets <code>nmap -data-length [size] <Target IP></code>
<code>--ip-options <options></code>	Send packets with specified IP options
<code>--ttl <val></code>	Set IP time-to-live field
<code>--spoof-mac <mac address/prefix/vendor name></code>	Spoof your MAC address <code>nmap -spoof-mac [MAC 0 vendor] <Target IP></code>
<code>--badsum</code>	Send packets with a bogus TCP/UDP/SCTP checksum

<code>--proxies url1,[url2],...</code>	Relay connections through HTTP/SOCKS4 proxies
OUTPUT	
<code>-oN/-oX/-oS/-oG <file></code>	Output scan in normal, XML, s <rlpt klddi3, and Grepable format, respectively, to the given filename
<code>-oA <basename></code>	Output in the three major formats at once
<code>-v</code>	Increase verbosity level (use -vv or more for greater effect) <code>nmap -v <Target IP></code>
<code>-d</code>	Increase debugging level (use -dd or more for greater effect) <code>nmap -d <Target IPs></code>
<code>--reason</code>	Display the reason a port is in a particular state
<code>--open</code>	Only show open (or possibly open) ports <code>nmap --open <Target IP></code>
<code>--packet-trace</code>	Show all packets sent and received <code>nmap --packet-trace <Target IP></code>
<code>--iflist</code>	Print host interfaces and routes (for debugging) <code>nmap --iflist</code>
<code>--log-errors</code>	Log errors/warnings to the normal-format output file
<code>--append-output</code>	Append to rather than clobber specified output files
<code>--resume <filename></code>	Resume an aborted scan
<code>--noninteractive</code>	Disable runtime interactions via keyboard
<code>--stylesheet <path/URL></code>	XSL stylesheet to transform XML output to HTML
<code>--webxml</code>	Reference stylesheet from Nmap.Org for more portable XML
<code>--no-stylesheet</code>	Prevent associating of XSL stylesheet w/XML output
<code>-stats-every [time]</code>	Periodically display statistics <code>nmap -stats-every [time] <Target IP></code>

Miscellaneous Options	
-h	Nmap help screen <code>nmap -h</code>
-6	IPv6 Scanning by using -6 option in Zenmap <code>nmap -6 scanme.nmap.org</code> Enable IPv6 scanning <code>nmap -6 2607:f0d0:1002:51::4</code> OS discovery using IPv6 fingerprinting method <code>nmap -6 -O <Target IP></code>
-A	Enables OS detection, version detection, script scanning, and traceroute, also known as Aggressive scan <code>nmap -A <Target IP></code>
-n	Disable reverse IP address lookups
--datadir <dirname>	Specify custom Nmap data file location
--send-eth/--send-ip	Send using raw ethernet frames or IP packets
--privileged	Assume that the user is fully privileged
-v	Display Nmap version <code>nmap -v</code>
--unprivileged	Assume the user lacks raw socket privileges

2. Nmap Port Scan types

Command	Description
<code>nmap -sT <Target IP></code>	Connect Scan (Default without root privileges)/ Scan using TCP connect
<code>nmap -sS <Target IP></code>	Scan using TCP SYN scan (default)
<code>nmap -sU <Target IP></code>	UDP Scan
<code>nmap -sA <Target IP></code>	ACK Scan
<code>nmap -sM <Target IP></code>	Maimon Scan

<code>nmap -sL <Target IP></code>	No Scan, list targets only
<code>nmap -sL -v <Target IP></code>	List scan
<code>nmap -Pn <Target IP></code>	Disable host discovery, port scanning
<code>nmap -PSx <Target IP></code>	SYN Discovery on port x, port 80 by default
<code>nmap -PUx <Target IP></code>	UDP discovery on port x, port 40125 by default
<code>nmap -PAx <Target IP></code>	ACK discovery on port x, port 80 by default
<code>nmap -PR <Target IP>/24</code>	ARP discovery on local network
<code>nmap -n <Target IP></code>	Never do DNS resolution
<code>nmap -p x <Target IP></code>	Scan for port x
<code>nmap -p 21-50 <Target IP></code>	Port Range
<code>nmap -p U:53,T:21-25,80</code>	Scan multiple TCP and UDP ports
<code>nmap -p- <Target IP></code>	Scan all ports
<code>nmap -p http,ftp <Target IP></code>	Port scan from service name
<code>nmap -F <Target IP></code>	Fast port scan (100 ports)
<code>nmap -f <Target IP></code>	Scan fragmented IP packets
<code>nmap --mtu x <Target IP></code>	Set own offset size x
<code>nmap --top-ports x <Target IP></code>	Scan the top x ports
<code>nmap -sV--version-intensity 5 <Target IP></code>	Aggressive service discovery
<code>nmap -sV --version-intensity 0 <Target IP></code>	Light banner grabbing
<code>nmap -sV--version-light <Target IP></code>	Enable light mode, lower possibility of correctness
<code>nmap -sV--version-all <Target IP></code>	Enable intensity level 9. Higher possibility of correctness

<code>nmap -O--osscan-limit <Target IP></code>	Limit OS detection to promising targets
<code>nmap -O--osscan-guess <Target IP></code>	Guess OS detection results
<code>nmap -O --max-os-tries x <Target IP></code>	Set maximum number of OS detection tries against a target
<code>nmap -sU -p 123,161,162 <Target IP></code>	Scan UDP ports
<code>nmap -Pn -F <Target IP></code>	Scan selected ports - ignore discovery
<code>nmap -Pn -sT --scan-delay 1s --max-parallelism 1 -p <Port List> <Target IP></code>	Identify open ports and services
<code>nmap -Pn -sT -p 46824 <Target IP></code>	Identify HMI systems
<code>nmap -Pn -sT -p 102 --script s7-info <Target IP></code>	Scan Siemens SIMATIC S7 PLCs
<code>nmap -Pn -sT -p 502 --script modbus-discover <Target IP></code>	Scan Modbus Devices
<code>nmap -sU -p 500 <Target IP></code>	Check the status of isakmp over port 500
<code>nmap -Pn -sU -p 47808 --script bacnet-info <Target IP></code>	ScanBACnet Devices
<code>nmap -Pn -sU -p 44818 --script enip-info <Target IP></code>	Scan Ethernet/IP Devices
<code>nmap -Pn -sT -p 1911,4911 --script fox-info <Target IP></code>	Scan Niagara Fox Devices
<code>nmap -Pn -sT -p 20547 --script proconos-info <Target IP></code>	Scan ProConOS Devices
<code>nmap -Pn -sT -p 9600 --script omron-info <Target IP></code>	Scan Omron PLC Devices
<code>nmap -Pn -sU -p 9600 --script omron-info <Target IP></code>	Scan Omron PLC Devices
<code>nmap -Pn -sT -p 1962 --script pcworx-info <Target IP></code>	Scan PCWorx Devices

3. Nmap Commands

Command	Description
<code>nmap -p 1-65535 -T4 -A -v <Target IP></code>	Perform intense scan on all TCP ports
<code>nmap -p ports <Target IP></code>	Run Nmap to identify IoT devices using insecure HTTP ports for transmitting data
<code>nmap -T4 -A -v -Pn <Target IP></code>	Perform Intense scan with no ping
<code>nmap -T4-A-v-PE-PS-PA Ports URL</code>	Footprint Web Infrastructure: Service Discovery
<code>nmap -sn <Target IP></code>	Perform ping scan
<code>nmap -sn <Target IP/Subnet></code>	Disable port scanning, host discovery only
<code>nmap -sn -PR <Target IP></code>	ARP Ping Scan
<code>nmap -sn -PU <Target IP></code>	UDP Ping Scan
<code>nmap -sn -PE <Target IP></code>	ICMP ECHO Ping Scan
<code>nmap -sn -PE <IP range></code>	ICMP ECHO Ping Sweep
<code>nmap -sn -PP <Target IP></code>	ICMP Timestamp Ping Scan
<code>nmap -sn -PM <Target IP></code>	ICMP Address Mask Ping Scan
<code>nmap -sn -PS <Target IP></code>	TCP SYN Ping Scan
<code>nmap -sn -PA <Target IP></code>	TCP ACK Ping Scan
<code>nmap -sn -PO <Target IP></code>	IP Protocol Ping Scan
<code>nmap -St -v <Target IP></code>	TCP Connect/ Full Open Scan
<code>nmap -sS -v <Target IP></code>	Stealth Scan (Half-open Scan)
<code>nmap -sX -v <Target IP></code>	Xmas Scan
<code>nmap -sM -v <Target IP></code>	TCP Maimon Scan
<code>nmap -sA -v <Target IP></code>	ACK Flag Probe Scan

<code>nmap -badsum <Target IP></code>	Sending Bad Checksums
<code>nmap --script smb-os-discovery.nse <Target IP></code>	OS Discovery using Nmap Script Engine
<code>nmap -sV -T4 -O -F -version-light <Target IP></code>	Perform quick scan plus
<code>nmap -sV -T4 -O -F -version-light scanme.nmap.org <Target URL></code>	Wi-Fi vulnerability scanning on wireless networks
<code>nmap -sV -O -p <Target IP></code> <code>nmap -sV --script http-enum <Target IP></code> <code>nmap target IP address -p 80 --script = http-frontpage-login</code> <code>nmap --script http-passwd --script-args http-passwd.root =/ <Target IP></code>	NSE scripts to enumerate information about the target website/ web servers
<code>nmap -sV --script http-enum <Target domain></code>	Analyze Web Applications: Identify exposed Files and Directories of the target webserver
<code>nmap -iL list-of-ips.txt</code>	Scan targets from a text file
<code>nmap --script=sniffer-detect [Target IP Address/Range of IP addresses]</code>	Command to detect NIC in promiscuous mode
<code>nmap <Target IP> --data 0xdeadbeef</code>	Create Custom Packets by Appending Custom Binary Data
<code>nmap <Target IP> --data-string "ph34r my 33t skills"</code>	Create Custom Packets by Appending Custom String
<code>nmap <Target IP> --data-string 5</code>	Create Custom Packets by Appending Random Data
<code>nmap -sU -p 500 <Target IP></code>	Perform a check on the status of ISAKMP over port 500
<code>nmap -sR <Target IP/network></code>	Identify the RPC service running on the network
<code>nmap --script hostmap <host></code>	Discover virtual domains with hostmap
<code>nmap --script http-trace -p80 localhost</code>	Detect a vulnerable server that uses the TRACE method
<code>nmap --script http-google-email <host></code>	Harvest email accounts with http-google-email
<code>nmap -p80 --script http-userdir -enum localhost</code>	Enumerate users with http-userdir-enum

<code>nmap -p80 --script http-trace <host></code>	Detect HTTP TRACE
<code>nmap -p80 --script http-waf-detect - -script-args="http-wafdetect. uri=/testphp.vulnweb.com/artists.php ,http-wafdetect. detectBodyChanges" www.modsecurity.org</code>	Check if web server is protected by WAF/IPS
<code>nmap --script http-enum -p80 <host></code>	Enumerate common web applications
<code>nmap -p80 --script http-robots.txt <host></code>	Obtain robots.txt
<code>nmap -p80 --script http-test.txt <host></code>	Obtain test.txt
<code>nmap --script=asn-query,whois,ip- geolocation-maxmind <Target IP/ Subnet></code>	IP address Information
<code>nmap --script=http-title <Target IP/ Subnet></code>	Gather page titles from HTTP services
<code>nmap --script=http-headers <Target IP/ Subnet></code>	Get HTTP headers of web services
<code>nmap --script=http-enum <Target IP/ Subnet></code>	Find web apps from known paths
<code>nmap -n -Pn -sSU -pT:0-65535,U:0- 65535 -v -A -oX <Name><Target IP></code>	Perform complete scan of the IoT device that checks for both TCP and UDP services and ports
<code>nmap -sS -T4 -A -f -v <Target IP></code>	Packet Fragmentation/ SYN/FIN scan using Nmap
<code>nmap -g 80 <Target IP></code>	Source Port Manipulation/ Use given source port number
<code>nmap -sU -A -PN -n -pU:19,53,123,161 -script=ntp-monlist,dns- recursion,snmp-sysdescr <Target IP/ network></code>	Scan for UDP DDOS reflectors
<code>nmap -6 -n -Pn -sSU -pT:0-65535,U:0- 65535 -v -A -oX <Name><Target IP></code>	Identify the IPv6 capabilities of a device
<code>nmap -T4 -A -v <Target IP></code>	Perform intense scan
<code>nmap -T4 -A <Target IP/Subnet></code>	Identify vulnerable services on service port by attackers by using RPC Enumeration
<code>nmap -p 23 <Target Domain></code>	Telnet Enumeration

<code>nmap -p 23 --script telnet-ntlm-info <Target IP></code>	Enumerate information from remote Microsoft Telnet services with NTLM authentication enabled
<code>nmap -p 23 -script telnet-brute.nse --script-args</code>	Perform brute-force attack against telnet server
<code>nmap -p 445 -A <Target IP></code>	Enumerate SMB service running on the target IP address/ SMB Enumeration
<code>nmap -p 21 <Target Domain></code>	FTP Enumeration
<code>nmap -p 69 <Target Domain></code>	Enumerate TFTP service running on the target domain
<code>nmap -p 179 <Target IP></code>	BGP Enumeration
<code>nmap -sS -sU -T4 -A -v <Target IP></code>	Perform intense scan and scanning for UDP
<code>nmap -sV -v -p 139,445 <Target IP/Subnet></code>	Detect all exposed Netbios servers on the subnet
<code>nmap -sV -v --script nbstat.nse <Target IP></code>	Nmap's nbstat NSE script allow attackers to retrieve target's NetBIOS names and MAC addresses
<code>nmap -sU --script nbstat.nse -p 137 <Target IP address></code>	Find target Netbios name
<code>nmap --script-args=unsafe=1 --script smb-check-vulns.nse -p 445 <Target IP address></code>	Check if Netbios servers are vulnerable to MS08-067
<code>nmap -sV --version-intensity 0 <Target IP></code>	Lighter banner grabbing detection
<code>nmap -sV --version-intensity 5 <Target IP></code>	More aggressive Service Detection
<code>nmap -sV <Target IP></code>	Attempts to determine the version of service running/ Standard service detection/ Service Version Discovery in Zenmap
<code>nmap --script-help=ssl-heartbleed</code>	Get help for a script
<code>nmap --script dns-zonetransfer.nse - -script-args dns- zonetransfer.domain=<domain> -p53 <hosts></code>	Attempts to pull a zone file (AXFR) from a DNS server
<code>nmap --script http-robots.txt <hosts></code>	Harvests robots.txt files from discovered web servers

<code>nmap --script smb-brute.nse -p445 <hosts></code>	Attempts to determine valid username and password combinations via automated guessing
<code>nmap --script smb-psexec.nse -script-args=smbuser=<username>, smbpass=<password>[, config=<config>] -p445 <hosts></code>	Attempts to run a series of programs on the target machine, using credentials provided as scriptargs
<code>nmap -sV -p 443 --script=ssl-heartbleed <Target IP/Subnet></code>	Detect Heartbleed SSL Vulnerability
<code>nmap <Target IP>-50 -sL --dns-server <Target IP></code>	Query the Internal DNS for hosts, list targets only
<code>nmap -iR 10 -sn -traceroute</code>	Traceroute to random targets, no port scan
<code>nmap <Target IP>-1/24 -PR -sn -vv</code>	Arp discovery only on local network, no port scan
<code>nmap -iR 10 -PS22-25,80,113,1050,35000 -v -sn</code>	Discovery only on ports x, no port scan
<code>nmap -sP <Target IP/Subnet></code>	Ping scans the network, listing machines that respond to ping
<code>nmap -v -sS -A -T4 <Target IP></code>	Prints verbose output, runs stealth syn scan, T4 timing, OS and version detection, traceroute and scripts against target services
<code>nmap -v -sV -O -sS -T5 <Target IP></code>	Prints verbose output, runs stealth syn scan, T5 timing, OS and version detection
<code>nmap -iL ip-addresses.txt</code>	Scans a list of IP addresses
<code>nmap --script-args=unsafe=1 --script smb-check-vulns.nse -p 445 <Target IP></code>	Check if Netbios servers are vulnerable to MS08-067
<code>nmap -Pn -p- -sI zombie target</code>	Attack
<code>nmap -b ftp rely host</code>	FTP Bounce Scan <username>:<password>@<server>:<port>. <Server> is the name or IP address of a vulnerable FTP server
<code>nmap -T0 <Target IP></code>	Paranoid (0) Intrusion Detection System evasion
<code>nmap -T1 <Target IP></code>	Sneaky (1) Intrusion Detection System evasion
<code>nmap -T2 <Target IP></code>	Polite (2) slows down the scan to use less bandwidth and use less target machine resources

<code>nmap -T3 <Target IP></code>	Normal (3) default speed
<code>nmap -T4 <Target IP></code>	Aggressive (4) speeds scan; assumes you are on a reasonably fast and reliable network
<code>nmap -T5 <Target IP></code>	Insane (5) speeds scan; assumes you are on extraordinarily fast network
<code>nmap --script=ftp <Target IP></code>	Scan with a single script
<code>nmap --script=http* <Target IP></code>	Scan with a wildcard script
<code>nmap --script=banner,http <Target IP></code>	Scan with two scripts
<code>nmap --script "not intrusive" <Target IP></code>	Scan default, but remove intrusive scripts
<code>nmap -Pn --script=http-sitemap-generator xyz.com</code>	HTTP site map generator
<code>nmap -n -Pn -p 80 --open -sV -vvv --script banner,http-title -iR 1000</code>	Fast search for random web servers
<code>nmap -Pn --script=dns-brute xyz.com</code>	Brute forces DNS hostnames guessing subdomain
<code>nmap -n -Pn -vv -O -sV --script smb-enum*,smb-ls,smb-mbenum,smb-os-discovery,smb-s*,smb-vuln*,smbv2* -vv <Target IP></code>	Safe SMB scripts to run
<code>nmap --script whois*<Target Domain></code>	Whois query
<code>nmap -p80 --script http-unsafe-output-escaping <Target Website></code>	Detect cross site scripting vulnerabilities
<code>nmap -p80 --script http-sql-injection <Target Website></code>	Check for SQL injections
<code>nmap --data-length x <Target IP></code>	Appends random data to sent packets
<code>nmap -oN file.file --append-output <Target IP></code>	Append a scan to a previous scan file
<code>nmap --iflist</code>	Shows the host interface and routes
<code>nmap -6 2607:f0d2:5664:51::5</code>	Enable IPV6 scanning
<code>nmap -T0 -b username:password@ftpserver.tld:21 victim.tld</code>	Uses the username "username", the password "password", the FTP server "ftpserver.tld" and port 21 on said server to scan victim.tld.

<code>nmap -sU -sT -p U:[ports],T:[ports] <Target IP></code>	Scan ports by protocol
<code>nmap -sV -version-trace <Target IP></code>	Troubleshooting version scans
<code>nmap -script [script.nse] <Target IP></code>	Execute individual scripts
<code>nmap -script [expression] <Target IP></code>	Execute multiple scripts
<code>nmap -script [category] <Target IP></code>	Execute scripts by category
<code>nmap -script [category1,category2, etc]</code>	Execute multiple scripts categories
<code>nmap -script [script] -script-trace <Target IP></code>	Troubleshoot scripts
<code>\$ docker -H <docker host> run --network=host --rm marsmensch/nmap -ox <IP Range></code>	Use Nmap to scan the host's internal network to identify running services
<code>ndiff [scan1.xml] [scan2.xml]</code>	Comparison using Ndiff
<code>ndiff -v [scan1.xml] [scan2.xml]</code>	Ndiff verbose mode
<code>ndiff -xml [scan1.xml] [scan2.xml]</code>	XML output mode

Port Selection

Command	Description
<code>nmap <Target IP></code>	Scan single IP
<code>nmap <Target IP> <Target IP></code>	Scan specific IPs
<code>nmap <Target IP range></code>	Scan a range of IPs
<code>nmap <Target Website></code>	Scan a host
<code>nmap <Target Domain></code>	Scan a domain
<code>nmap <Target IP/Subnet></code>	Scan using CIDR notation
<code>nmap -iL file.txt</code>	Scan targets using given file
<code>nmap --exclude <Target IP></code>	Exclude listed host/ specified IP s exclude from scan
<code>nmap -iR 50</code>	Scan 50 random hosts

NSE Scripts	
<code>nmap -sC <Target IP></code>	Scan with default NSE scripts.
<code>nmap --script-default <Target IP></code>	Scan with default NSE scripts.
<code>nmap --script snmp-sysdescr --script-args snmpcommunity=admin <Target IP></code>	NSE script with arguments
<code>nmap -script-args-file=filename</code>	Provide NSE script args in a file
<code>nmap -sV -sC <Target IP></code>	Scan using default safe scripts
<code>nmap -sV --script=smb* <Target IP></code>	Scan with a set of scripts
<code>nmap -sV -p 443 -script=ssl-heartbleed.nse <Target IP></code>	Scan using a specific NSE script